

S3KG Benchmarking

What we measured, why it matters, and what the results show

1. What is this benchmarking about?

The poster's benchmarking section evaluates **S3KG**, our proposed semantic similarity method that combines Knowledge Graph (KG) structure with sentence transformer embeddings against seven standard baselines across nine datasets. The central question is:

Does adding KG structural information actually improve semantic similarity detection, and does it generalise across different text types?

2. The seven baselines, what are they?

We compare against three categories of baseline:

Lexical overlap methods (surface-level)

- **ROUGE-1 / ROUGE-2 / ROUGE-L**: Count word (unigram, bigram) or longest-sequence overlaps between two texts. Fast but blind to meaning, two sentences can share all the same words yet mean opposite things.
- **BLEU**: Originally designed for machine translation quality; measures n-gram precision. Similar limitations to ROUGE.

Token-level neural method

- **BERTScore**: Uses BERT embeddings to match tokens between two texts, giving soft overlap rather than hard word counts. More robust than ROUGE/BLEU but still operates at the token level.

Dense sentence embedding models

- **MiniLM**: A compact transformer fine-tuned for semantic sentence similarity. Encodes each sentence as a single vector and compares cosine distance.
- **sentence-T5-base (sent-T5)**: A T5-based sentence encoder also fine-tuned for semantic tasks. The strongest baseline overall, this is our main neural competitor.

3. The nine benchmark datasets

Datasets span three structural categories, each testing a different capability:

Short text - human annotated (3 datasets)

- **MRPC**: News sentence pairs (avg. 10 words) with human paraphrase labels.
- **PAWS-Wiki**: Adversarially constructed pairs where word overlap is deliberately misleading — nearly identical surface form but different meaning, or vice versa. Hard for lexical methods.
- **STS12**: Sentence similarity pairs drawn from multiple NLP tasks.

KG-perturbed paragraphs (5 datasets)

Longer paragraphs (avg. 70–126 words) where entity relationships from a Knowledge Graph are deliberately swapped to create semantically different but surface-similar text pairs. The five datasets differ by domain:

- **SK-Codex 400 (C400):** General encyclopaedic knowledge (Codex KG).
- **SK-Combined (Comb.):** Mixed across multiple KG domains.
- **SK-FindKG (Find):** Financial and economic domain, noisy entity extraction.
- **SK-GloBI (GloBI):** Biological interaction triples, highly structured relationships.
- **SK-Oregano (Oreg.):** Food ontology domain.

Anti-circularity control (1 dataset)

- **Wikipedia Entity-Swap (W-Swap):** Entities swapped independently of any KG pipeline. This is a critical test: if S3KG's gains were just because our test data was built by the same pipeline that trains the method (circularity), performance should collapse here. It doesn't, confirming genuine generalisation.

4. What does S3KG do differently?

S3KG extracts Knowledge Graph triples from text (subject, relation, object) and converts them into a structural embedding. It then *interpolates* this KG structural signal with a standard sentence transformer using a mixing coefficient α :

$$\text{sim}_\alpha = (1 - \alpha) \times \text{sim_KG} + \alpha \times \text{sim_ST}$$

- $\alpha = 0.0$: Pure KG structural embedding - captures relational structure but lacks lexical range.
- $\alpha = 1.0$: Pure sentence transformer - strong semantics but blind to graph topology.
- **Best α :** Swept from 0.0 to 1.0 per dataset; the optimal value differs by domain.

The key insight: KG triples capture **directed relationships** between entities. If two sentences describe the same event with reversed subject/object roles (e.g., 'Furlan beat Johansson' vs. 'Johansson beat Furlan'), ROUGE scores both pairs identically, all tokens match. S3KG detects the structural reversal.

5. F1 score results

Darker blue = higher F1. Bolded cells mark the best-performing method per dataset column.

	Short Text			KG-Perturbed Paragraphs					Anti-C.
Method	MRPC	PAWS	STS12	C400	Comb.	Find	GloBI	Oreg.	W-Swap
S3KG	0.69	0.77	0.79	0.93	0.83	0.77	0.89	0.81	0.87
ROUGE-1	0.74	0.68	0.72	0.83	0.73	0.74	0.78	0.74	—
ROUGE-2	0.72	0.71	0.68	0.82	0.71	0.72	0.78	0.75	0.86
ROUGE-L	0.73	0.73	0.70	0.79	0.72	0.72	0.76	0.79	0.73
BLEU	0.69	0.72	0.67	0.81	0.71	0.71	0.78	0.74	0.87
BERTScore	0.76	0.69	0.68	0.82	0.76	0.74	0.82	0.74	0.75

MiniLM	0.72	0.69	0.83	0.88	0.77	0.80	0.78	0.77	0.82
sent-T5	0.77	0.67	0.85	0.88	0.77	0.85	0.73	0.80	0.76

6. AUROC results

AUROC measures discrimination ability across all thresholds, less sensitive to threshold choice than F1.

	Short Text			KG-Perturbed Paragraphs					Anti-C.
Method	MRPC	PAWS	STS12	C400	Comb.	Find	GloBI	Oreg.	W-Swap
S3KG	0.67	0.80	0.83	0.97	0.83	0.80	0.94	0.89	0.89
ROUGE-1	0.78	0.49	0.75	0.92	0.73	0.74	0.83	0.78	—
ROUGE-2	0.72	0.72	0.66	0.89	0.71	0.71	0.83	0.79	0.77
ROUGE-L	0.76	0.81	0.71	0.85	0.72	0.72	0.80	0.83	0.31
BLEU	0.68	0.75	0.64	0.88	0.71	0.71	0.82	0.79	0.74
BERTScore	0.82	0.70	0.64	0.92	0.79	0.76	0.87	0.80	0.65
MiniLM	0.75	0.64	0.89	0.94	0.79	0.84	0.82	0.81	0.81
sent-T5	0.82	0.67	0.93	0.94	0.83	0.90	0.76	0.87	0.81

7. Key findings to communicate to evaluators

S3KG leads on 7 of 9 datasets

In both F1 and AUROC, S3KG is the top method on 7 out of 9 datasets. The two exceptions are MRPC (very short text, sparse KG triples) and SK-FindKG (noisy financial vocabulary degrades KG extraction).

Biggest gains on KG-perturbed paragraphs

- **SK-Codex 400:** +5.7 F1 over best baseline (MiniLM 0.875 → S3KG 0.932)
- **SK-Combined:** +6.4 F1 over sent-T5 (0.770 → 0.834)
- **SK-GloBI:** +7.6 F1 over BERTScore (0.816 → 0.892) - biological triples are highly structured, which is where S3KG excels most

Performance scales with text length and relational richness

Short texts (10–22 words) produce sparse KG triples, limiting S3KG's advantage. Paragraph-length texts (70–126 words) yield richer graphs and larger margins. Performance also correlates with how distinctive the KG domain's relations are: GloBI (biological interactions) > Codex (encyclopaedic) > FindKG (noisy financial).

ROUGE-1 collapses on adversarial datasets

- **PAWS-Wiki:** ROUGE-1 AUROC = 0.490 (near-random). Lexical overlap is useless when word order is deliberately deceptive.

- **Wiki-Swap:** ROUGE-L AUROC = 0.311 (below random). ROUGE-1 appears perfect but only due to a dataset artefact (token overlap is trivially high by construction), so it is excluded from meaningful comparison on this dataset.

Anti-circularity confirmed

On Wiki-Swap, constructed entirely independently of our KG pipeline, S3KG scores $F1 = 0.872$, $AUROC = 0.890$, beating all valid baselines (next best: ROUGE-2 at $F1 = 0.860$, $AUROC = 0.772$; sent-T5 at $F1 = 0.762$). This rules out the concern that our results are an artefact of circular evaluation.

sent-T5 is the closest competitor but collapses on relational datasets

sent-T5 outperforms S3KG on MRPC, STS12, and SK-FindKG (where KG extraction is noisy), and is competitive on SK-Oregano. However, it drops sharply on SK-GloBI ($F1 = 0.728$, vs S3KG 0.892) and Wiki-Swap ($F1 = 0.762$ vs S3KG 0.872), confirming that dense embeddings alone cannot capture relational structure.

8. One-line verdict per dataset

Dataset	Verdict	Why
MRPC	Behind	Very short text → sparse KG triples. Dense models win.
PAWS-Wiki	Strong (+3.1 F1)	Adversarial paraphrases expose ROUGE/BLEU failure. KG structure discriminates correctly.
STS12	Behind	Short text; dense embeddings are better calibrated here.
SK-Codex 400	Strong (+5.7 F1)	Well-formed encyclopaedic triples; structural signal is clean and discriminative.
SK-Combined	Strong (+6.4 F1)	Mixed domains still favour structural approach; generalises beyond single KG.
SK-FindKG	Behind (domain noise)	Financial/economic vocabulary confuses KG extraction; noisy triples hurt.
SK-GloBI	Strong (+7.6 F1)	Biological interaction triples are highly distinctive; biggest S3KG gain.
SK-Oregano	Comparable (+1.2 F1)	Food ontology is moderately structured; small but consistent advantage.
Wiki-Swap	Best meaningful score	Independent construction rules out circularity. S3KG generalises.

Note for evaluators: All datasets are balanced at 400 pairs (200 positive, 200 negative). Performance is reported as maximum F1 via threshold sweep and AUROC. For S3KG, the best-performing α is selected per dataset from a sweep $\alpha \in \{0.0, 0.1, \dots, 1.0\}$.